

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#installing necessary packages
install.packages("tidyverse")
install.packages("lubridate")
install.packages("here")

#loading packages into library
library(tidyverse)
library(lubridate)
library(here)

#checking working directory
getwd()
```

```
## [1] "/home/guest/ENV 872"
```

```
here()
```

```
## [1] "/home/guest/ENV 872"
```

```
#loading datasets
```

```
Neonics <- read.csv(  
  file = here("Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),  
  stringsAsFactors = TRUE)  
  
Litter <- read.csv(  
  file = here("Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),  
  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Neonicotinoids are a type of insecticides, meaning they are applied to agriculture sites to control unwanted pests. However, they also have been shown to effect non-target insects like bees in negative ways, meaning close monitoring of their application and ecotoxicology is key to protect certain insect groups from being harmed. Additionally, understanding the effect of this insecticide on target insects in relation to their age and species could be of interest to realize the effect of the chemical based on life-history traits and exposure types.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Forest litter and woody debris likely play a critical role in the cycling of organic matter in forest environments. Additionally, the presence of forest litter could determine risks of the forest to various natural disasters, such as wildfire threats. Forest litter also can serve as a proxy for terrestrial net primary productivity.

4. How is litter and woody debris sampled as part of the NEON network? Read the `NEON_Litterfall_UserGuide.pdf` document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter and woody debris are weighed and categorized by plant functional groups, and they are collected with elevated and/or ground traps. These analyses only happen at sites with woody vegetation >2m tall. 2. Mass accuracy for dry weights is to 0.01 grams. Weights less than 0.01g are reported and represent the presence of a functional group, but not a reliable mass. 3. Masses are reported from a single trap for spatial resolution and from one collection event for temporal resolution.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#dimensions of dataset  
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#sorting the effect column from lowest to highest counts  
sort(summary(Neonics$Effect))
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)  
##           1           5           7           9  
##      Biochemistry      Accumulation      Intoxication      Immunological  
##           11           12           12           16  
##      Morphology      Growth      Enzyme(s)      Genetics  
##           22           38           62           82  
##      Avoidance      Development      Reproduction      Feeding behavior  
##          102          136          197          255  
##      Behavior      Mortality      Population  
##          360          1493          1803
```

Answer: The most common effects studied are population, mortality, and behavior. These would specifically be of interest because they are the most common responses from introduction of the insecticide, and they involve critical functions of the organism that represent how the chemical impacts them.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#summarizing the six most common species with the summary and maxsum function  
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee  
##           667           285           183  
##      Carniolan Honey Bee      Bumble Bee      Italian Honeybee  
##           152           140           113  
##      (Other)  
##          3083
```

Answer: Honey Bee, Parasitic Wasp, Buff Tailed Bumblebee, Carniolan Honey Bee, Bumble Bee, Italian Honeybee. These species are all stinging and likely pollinating insects, which the insecticides are not aiming to harm usually, so their impact is of interest.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
#determining data type for Conc.1..Author.
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

```
#seeing the number of levels within the column
levels(Neonics$Conc.1..Author.)
```

```
##      [1] "<0.0004"    "<0.025"     "<0.088"     "<0.5"        "<1.5"        "<10/"
##      [7] "<2.5/"      "<4.00"      "<5.00"      "<8.79"       ">=13.66"     ">0.01"
##     [13] ">0.1"       ">0.220"     ">0.5"       ">100.0"      ">1000.00"    ">20"
##     [19] ">2500.00"   ">5.0"       ">6300"      ">81"         ">98"         ">98.43"
##     [25] "~10"        "~30/"       "~40/"       "~41"         "0.000006"    "0.0000073"
##     [31] "0.000047"   "0.000048"   "0.00005"    "0.000069/"   "0.00007"     "0.000074"
##     [37] "0.00008"    "0.0000814"  "0.0001"     "0.000101"    "0.00017"     "0.0002"
##     [43] "0.0002/"    "0.00027/"   "0.00039"    "0.0004/"     "0.0005"      "0.0005/"
##     [49] "0.00053"    "0.0006"     "0.00071"    "0.00077"     "0.00084/"    "0.0009"
##     [55] "0.001"      "0.0011"     "0.0013"     "0.0014"      "0.0015"      "0.002"
##     [61] "0.0022/"    "0.0027"     "0.003"      "0.003/"      "0.0032"      "0.00322"
##     [67] "0.0035/"    "0.00355/"   "0.00368"    "0.0037/"     "0.0039"      "0.004"
##     [73] "0.005"      "0.005/"     "0.0056"     "0.0063"      "0.007"        "0.007/"
##     [79] "0.00739"    "0.0076"     "0.008"      "0.009"       "0.0095"      "0.01"
##     [85] "0.01/"      "0.010725"   "0.011"      "0.0112"      "0.0113"      "0.012"
##     [91] "0.012/"     "0.0125"     "0.013"      "0.013/"      "0.014"        "0.015"
##     [97] "0.016"      "0.017"      "0.0179"     "0.018"       "0.0192/"     "0.02"
##    [103] "0.02/"      "0.0206/"    "0.021"      "0.0218"      "0.022"        "0.023"
##    [109] "0.0231"     "0.024"      "0.025"      "0.025/"      "0.026"        "0.026/"
##    [115] "0.027"      "0.028"      "0.0298"     "0.0299"      "0.03"         "0.031"
##    [121] "0.032"      "0.033"      "0.035"      "0.037"       "0.0373"       "0.0375"
##    [127] "0.04"       "0.04/"      "0.0400/"    "0.0428"      "0.0429/"     "0.0439"
##    [133] "0.047"      "0.047/"     "0.048/"     "0.05"        "0.05/"        "0.051"
##    [139] "0.053"      "0.053/"     "0.056"      "0.056/"      "0.06"         "0.06/"
##    [145] "0.0600/"    "0.061"      "0.061/"     "0.0626"      "0.063"        "0.064"
##    [151] "0.0641"     "0.0671"     "0.06825/"   "0.07"         "0.07/"        "0.071"
##    [157] "0.075"      "0.077"      "0.078"      "0.08"        "0.08/"        "0.080/"
##    [163] "0.081"      "0.0812/"    "0.084/"     "0.085"       "0.088"        "0.09"
##    [169] "0.09/"      "0.091"      "0.1"        "0.1/"        "0.102"        "0.1039"
##    [175] "0.105"      "0.112"      "0.112/"     "0.113"       "0.114/"       "0.116/"
##    [181] "0.118"      "0.12"       "0.123"      "0.124"       "0.125"        "0.125/"
##    [187] "0.126/"     "0.13"       "0.13/"      "0.1332"      "0.134"        "0.137"
##    [193] "0.138"      "0.14"       "0.14/"      "0.141"       "0.145"        "0.15"
##    [199] "0.15/"      "0.150/"     "0.1500/"    "0.154"       "0.154/"       "0.155"
##    [205] "0.16"       "0.16/"      "0.161"      "0.17"        "0.17/"        "0.18"
##    [211] "0.18/"      "0.2"        "0.2/"       "0.202"       "0.21"         "0.21/"
##    [217] "0.2175"     "0.225"      "0.225/"     "0.227"       "0.23"         "0.233"
##    [223] "0.24"       "0.24/"      "0.246"      "0.25"        "0.25/"        "0.26"
##    [229] "0.26/"      "0.267"      "0.28"       "0.28/"       "0.29"         "0.29/"
##    [235] "0.291"      "0.295"      "0.297"      "0.3"         "0.3/"         "0.30/"
##    [241] "0.302"      "0.31"       "0.32"       "0.33/"       "0.336"        "0.34"
##    [247] "0.35"       "0.36"       "0.37"       "0.37/"       "0.375"        "0.38"
```

##	[253]	"0.39/"	"0.397"	"0.4"	"0.4/"	"0.40/"	"0.41"
##	[259]	"0.42"	"0.428"	"0.43"	"0.43/"	"0.44"	"0.4483"
##	[265]	"0.45"	"0.45/"	"0.46"	"0.46/"	"0.47"	"0.470/"
##	[271]	"0.48"	"0.481"	"0.49"	"0.49/"	"0.491"	"0.493"
##	[277]	"0.494"	"0.5"	"0.5/"	"0.50/"	"0.51"	"0.52"
##	[283]	"0.52/"	"0.53"	"0.536"	"0.558"	"0.56"	"0.56/"
##	[289]	"0.567/"	"0.57"	"0.58"	"0.580/"	"0.594"	"0.6"
##	[295]	"0.6/"	"0.609"	"0.61"	"0.625"	"0.63/"	"0.64"
##	[301]	"0.647"	"0.692"	"0.699/"	"0.7"	"0.71"	"0.72"
##	[307]	"0.72/"	"0.73"	"0.74/"	"0.745"	"0.75"	"0.75/"
##	[313]	"0.76/"	"0.79"	"0.8/"	"0.83"	"0.84"	"0.87"
##	[319]	"0.88"	"0.9"	"0.91"	"0.93"	"0.93/"	"0.946"
##	[325]	"0.95"	"0.9535"	"0.96"	"0.96/"	"0.98"	"1"
##	[331]	"1.0/"	"1.00/"	"1.01"	"1.04"	"1.051"	"1.07"
##	[337]	"1.08"	"1.09/"	"1.092"	"1.1"	"1.11"	"1.12"
##	[343]	"1.12/"	"1.16"	"1.2"	"1.20/"	"1.21"	"1.24"
##	[349]	"1.25"	"1.25/"	"1.28"	"1.28/"	"1.29"	"1.3"
##	[355]	"1.3/"	"1.34/"	"1.35"	"1.37"	"1.38"	"1.4"
##	[361]	"1.43/"	"1.44"	"1.45"	"1.48/"	"1.5"	"1.5/"
##	[367]	"1.51"	"1.51/"	"1.53"	"1.58"	"1.59"	"1.6/"
##	[373]	"1.62"	"1.64"	"1.65"	"1.67"	"1.69"	"1.7"
##	[379]	"1.72"	"1.75"	"1.75/"	"1.8"	"1.81"	"1.81/"
##	[385]	"1.82"	"1.847"	"1.86"	"1.93"	"1.99"	"1/"
##	[391]	"10"	"10.0/"	"10.3"	"10.37"	"10.5/"	"10.62"
##	[397]	"10/"	"100"	"100/"	"1000"	"1000/"	"1023"
##	[403]	"103.705"	"1034.8"	"104"	"104.1/"	"104/"	"105"
##	[409]	"105/"	"1056.7"	"1075.7"	"108.27"	"109.579"	"11.3"
##	[415]	"11.7"	"110.3"	"1104"	"112"	"112.5"	"115/"
##	[421]	"12"	"12.0/"	"12.02"	"12.17"	"12.429"	"12.5"
##	[427]	"12.5/"	"12.50/"	"12.57"	"12.8"	"12/"	"120"
##	[433]	"120.65"	"120/"	"1200/"	"122.4"	"124.03"	"125"
##	[439]	"125/"	"1267.9"	"127"	"13"	"13.3"	"13.3/"
##	[445]	"13.35"	"13.45"	"13.66"	"13.75"	"13.8"	"13.9"
##	[451]	"13.9/"	"132"	"133"	"134.80/"	"13407"	"136"
##	[457]	"136.25/"	"136/"	"137/"	"138.21"	"138.84"	"14"
##	[463]	"14.34"	"14.53"	"14.6"	"14.785"	"14/"	"1400"
##	[469]	"144"	"144.0/"	"144/"	"145.3"	"148.3"	"15"
##	[475]	"15.09"	"15.1"	"15.3"	"15.63"	"15/"	"150"
##	[481]	"150.46"	"150.5"	"150/"	"1500"	"1507"	"156"
##	[487]	"16"	"16.2"	"16.4"	"16.69"	"16/"	"167.9"
##	[493]	"168/"	"17"	"17.24"	"17.32"	"17.5"	"17.6"
##	[499]	"17.92"	"170"	"170.52"	"173.088"	"176.5"	"18"
##	[505]	"18.5"	"180"	"183.4"	"187.208"	"19.24"	"19.7"
##	[511]	"190.2"	"191.044"	"193.59"	"2"	"2.0/"	"2.008"
##	[517]	"2.1"	"2.15"	"2.16"	"2.17"	"2.18/"	"2.19"
##	[523]	"2.2"	"2.2/"	"2.217"	"2.24/"	"2.26"	"2.27/"
##	[529]	"2.3"	"2.31"	"2.38"	"2.4"	"2.403"	"2.403/"
##	[535]	"2.44"	"2.46"	"2.462"	"2.48"	"2.5"	"2.5/"
##	[541]	"2.556"	"2.6"	"2.6/"	"2.60/"	"2.608"	"2.61"
##	[547]	"2.63"	"2.644"	"2.667"	"2.68"	"2.689"	"2.75"
##	[553]	"2.761"	"2.78"	"2.79"	"2.8"	"2.82"	"2.84"
##	[559]	"2.844"	"2.85"	"2.86"	"2.89"	"2.9"	"2.936"
##	[565]	"2.94"	"2.947"	"2.96"	"2.967"	"2.99"	"2/"
##	[571]	"20"	"20.0/"	"20.3"	"20.4"	"20.6"	"20.636"

##	[577]	"20/"	"200"	"200/"	"203.6"	"2051"	"208.9"
##	[583]	"21.06"	"21.19"	"21.209"	"21.4"	"21.418"	"21.5"
##	[589]	"21.6"	"21/"	"212"	"217"	"218.89"	"22"
##	[595]	"22.2"	"22.4"	"22.59"	"222.65"	"224.05/"	"2240.5"
##	[601]	"225"	"228.8"	"23.37"	"23.37/"	"23.54"	"23.6"
##	[607]	"23.8"	"23.93"	"232.37"	"239.1"	"24"	"24.33"
##	[613]	"24.46"	"24.54"	"24.96"	"240/"	"241.3"	"242"
##	[619]	"242.45"	"245.5"	"249.23/"	"25"	"25.0/"	"25.39"
##	[625]	"25/"	"250"	"250/"	"251.3"	"251.3/"	"2558"
##	[631]	"26.025/"	"26.63"	"26.9"	"26/"	"263.44"	"2675"
##	[637]	"268"	"27.16"	"27.2"	"272.5/"	"276"	"28"
##	[643]	"28.2"	"28.4"	"28.81"	"28/"	"286/"	"288"
##	[649]	"29.79"	"296.62"	"3"	"3.0/"	"3.018"	"3.03"
##	[655]	"3.09"	"3.11"	"3.12"	"3.151"	"3.18"	"3.2"
##	[661]	"3.21"	"3.23"	"3.28"	"3.3"	"3.31"	"3.313"
##	[667]	"3.336"	"3.36"	"3.38"	"3.392"	"3.392/"	"3.4"
##	[673]	"3.41"	"3.42"	"3.4859"	"3.49"	"3.5"	"3.5/"
##	[679]	"3.53"	"3.56"	"3.59"	"3.6"	"3.62"	"3.68"
##	[685]	"3.7"	"3.743"	"3.747"	"3.75"	"3.75/"	"3.789"
##	[691]	"3.8"	"3.8/"	"3.82"	"3.885"	"3.9"	"3/"
##	[697]	"30"	"30.3"	"30.6"	"30.7"	"30/"	"300"
##	[703]	"31.16"	"31.3"	"31.37"	"31.42"	"311.9"	"32"
##	[709]	"32.4"	"32.8"	"32/"	"326.69"	"33.6"	"3390"
##	[715]	"34.3"	"34.37"	"34.96"	"35"	"35.183"	"35.813"
##	[721]	"350/"	"36/"	"360/"	"364.07"	"365.6"	"37"
##	[727]	"37.01"	"37.5/"	"37.83"	"376.3"	"382.31"	"383"
##	[733]	"39"	"39.32"	"39.37"	"4"	"4.0/"	"4.07"
##	[739]	"4.08"	"4.1"	"4.13"	"4.15"	"4.17"	"4.24"
##	[745]	"4.27"	"4.28"	"4.29"	"4.316"	"4.32"	"4.34"
##	[751]	"4.369"	"4.37"	"4.375/"	"4.383"	"4.39/"	"4.4"
##	[757]	"4.411"	"4.44"	"4.45"	"4.485"	"4.507"	"4.51"
##	[763]	"4.514"	"4.52"	"4.53"	"4.546"	"4.552"	"4.6"
##	[769]	"4.61"	"4.627"	"4.64"	"4.66"	"4.671"	"4.679"
##	[775]	"4.71"	"4.75"	"4.76"	"4.8"	"4.82"	"4.83"
##	[781]	"4.92"	"4.93"	"4/"	"40"	"40.0/"	"40.019"
##	[787]	"40.44"	"40.7"	"40/"	"400/"	"41.6"	"41.94"
##	[793]	"42"	"42.9"	"42.9/"	"43.02"	"43.4"	"43.59"
##	[799]	"43.7"	"4399"	"440"	"447.82"	"45.37"	"45.4/"
##	[805]	"45.9"	"450"	"451"	"46.7"	"46.763"	"46/"
##	[811]	"461.2"	"462.9"	"47"	"48"	"48.2"	"48.96"
##	[817]	"48/"	"480"	"49.42"	"5"	"5.0/"	"5.01"
##	[823]	"5.06"	"5.07"	"5.08"	"5.1"	"5.1/"	"5.101"
##	[829]	"5.11"	"5.16"	"5.18"	"5.2"	"5.25"	"5.27"
##	[835]	"5.28"	"5.3"	"5.3/"	"5.35"	"5.36"	"5.37"
##	[841]	"5.38"	"5.38/"	"5.4"	"5.46"	"5.49"	"5.53"
##	[847]	"5.56"	"5.6"	"5.64"	"5.7"	"5.71"	"5.73"
##	[853]	"5.77"	"5.79"	"5.8"	"5/"	"50"	"50.28"
##	[859]	"50/"	"500"	"500/"	"503.6"	"51.16"	"51.32"
##	[865]	"52.05/"	"52.2"	"52.6/"	"53"	"537"	"54.67"
##	[871]	"545/"	"56"	"56.35"	"56.73"	"57"	"57.7"
##	[877]	"5737"	"59"	"59.36"	"6"	"6.0/"	"6.03"
##	[883]	"6.13"	"6.135"	"6.25"	"6.27"	"6.36"	"6.5"
##	[889]	"6.55"	"6.57"	"6.6"	"6.7"	"6.7/"	"6.76"
##	[895]	"6.80/"	"6.83"	"6.95"	"6/"	"60"	"60.1"

```
## [901] "60.7"      "60/"      "600"      "61"       "62.5/"    "620.21"
## [907] "63.54"     "635.4"    "636.5"    "64.6"     "65.14"    "65.37"
## [913] "65.56"     "65.68"    "675.4"    "683.2"    "6830"     "695.5"
## [919] "7"         "7.07"     "7.14"     "7.32"     "7.379"    "7.379/"
## [925] "7.5"       "7.5/"     "7.6"      "7.7"      "7.72"     "7.8"
## [931] "7/"        "70"       "70/"      "71"       "71.3"     "71/"
## [937] "721.6"     "74.631"   "74.9"     "75"       "75.26"    "75.27"
## [943] "75/"       "750"      "754.2"    "76"       "76.035"   "76/"
## [949] "7839"      "788.5"    "8"        "8.0/"     "8.03"     "8.1"
## [955] "8.38"      "8.51"     "8.6"      "8.86"     "8/"       "80"
## [961] "80.66"     "80.9"     "80/"      "800"      "8000"     "80000"
## [967] "801"       "81.09"    "83"       "83.97"    "8352"     "84"
## [973] "84/"       "85.3"     "85.47"    "85.8"     "85.8/"    "86.9"
## [979] "87.1"      "87.5"     "9"        "9.07"     "9.49"     "9.5"
## [985] "9.5858"    "9.6"      "9.89"     "9.94"     "90"       "90/"
## [991] "900"       "93.21"    "94"       "95.48"    "95.8"     "96"
## [997] "96/"       "97.425"   "98.43"    "98.43/"   "99.063"   "99.75"
## [1003] "99.82"     "995"      "NR"       "NR/"
```

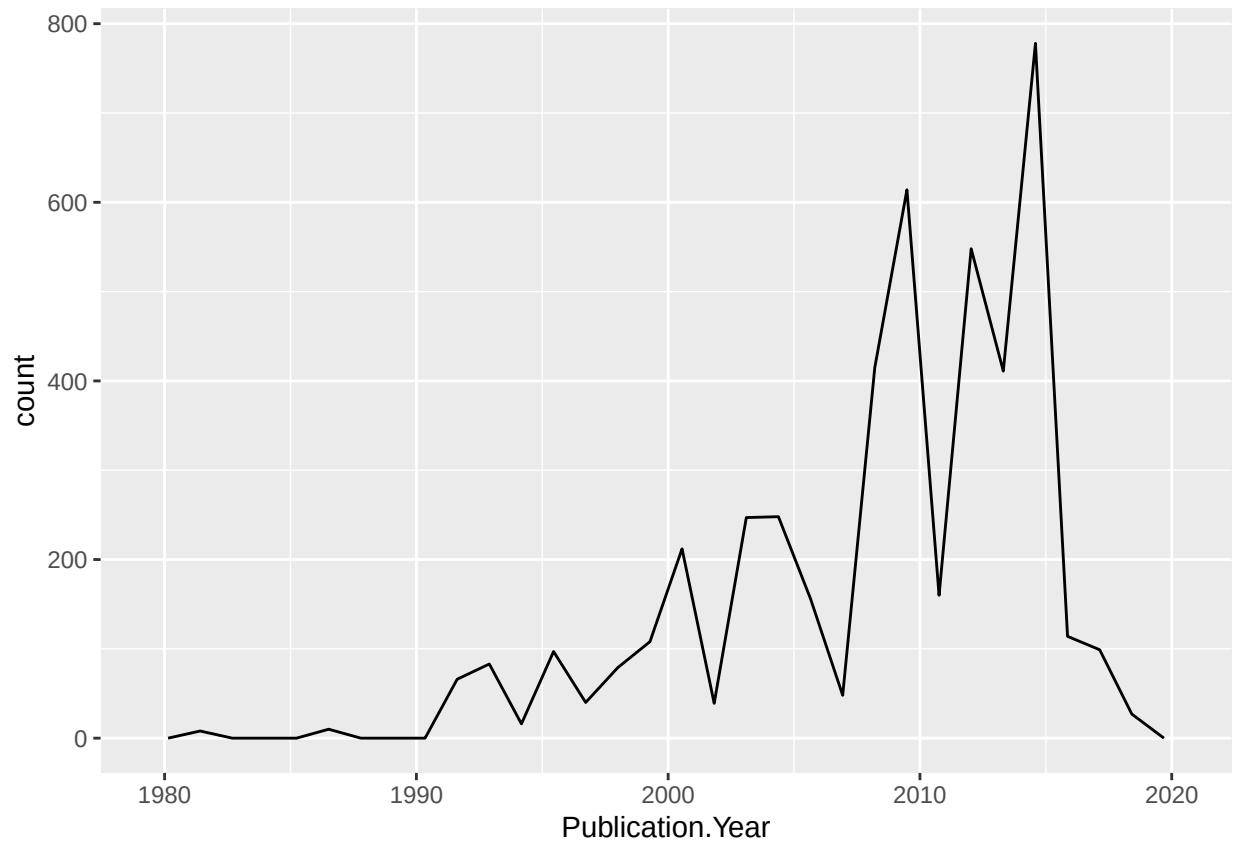
Answer: Factor. It is not numeric because some of the data inputs contain the “/” symbol, and numeric variables can only contain integers.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
#Plotting number of studies each publication year with ggplot
ggplot(Neonics, aes(Publication.Year, bins = 10)) + geom_freqpoly()
```

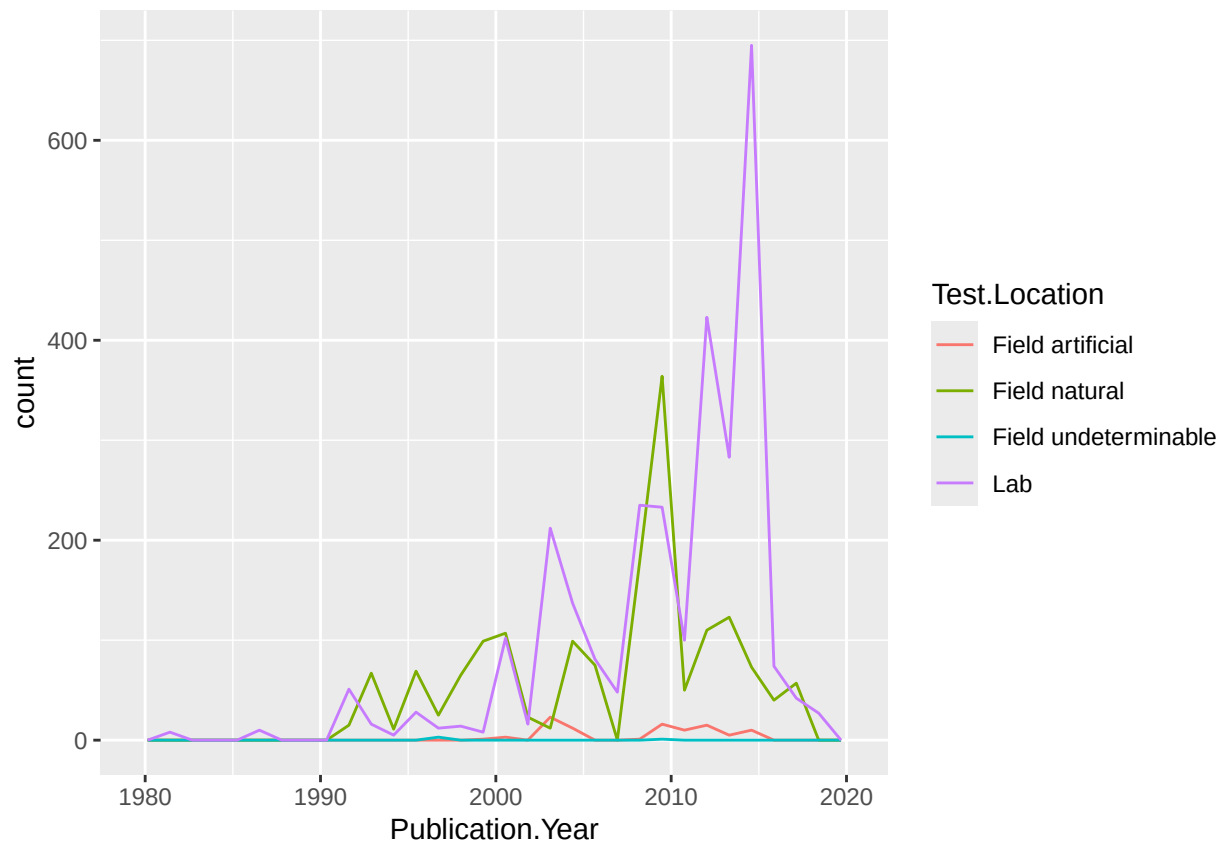
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
#Plotting number of studies each publication year with ggplot and
#indicating the test location by color
ggplot(Neonics, aes(Publication.Year, bins = 10, colour = Test.Location)) +
  geom_freqpoly()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

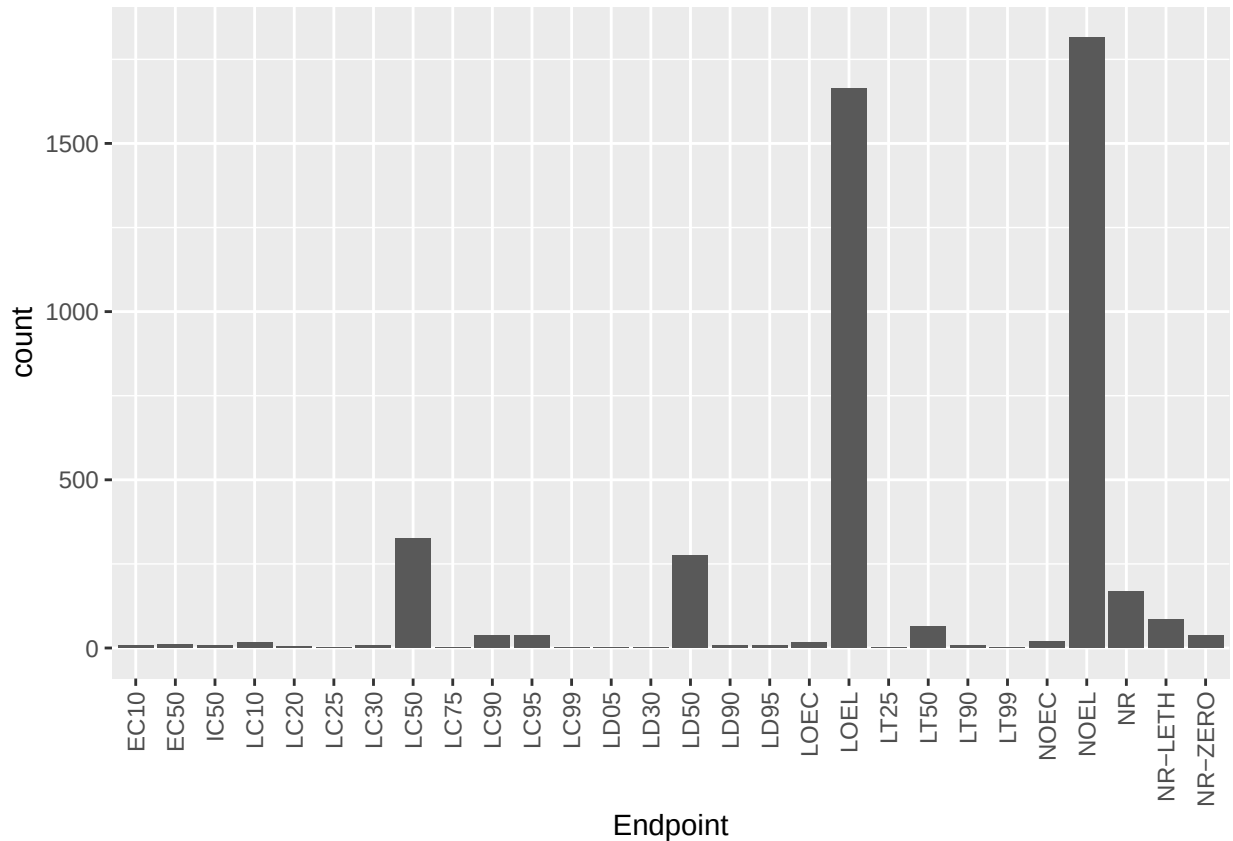
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are the lab and natural field, with artificial field and undeterminable field consistently the lowest. Natural field and lab do differ overtime, with both showing periods of being the most common test location.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
#Plotting counts of endpoints with a bar graph
ggplot(Neonics, aes(Endpoint)) + geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: Two most common endpoints are LOEL and NOEL. LOEL is a terrestrial “lowest observable effect level”: the lowest dose producing significantly different effects during experiments. NOEL is a terrestrial “no observable effect level”: highest dose producing no significant effects from the controls.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#determining class of collectDate
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#changing collectDate to a date
Litter$collectDate <- ymd(Litter$collectDate)
```

```
#confirming collectDate is now a date
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
#determine sampling dates from August 2018
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

Answer: Litter was sampled on August 2 and 30 of 2018.

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#listing plotIDs with unique function
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

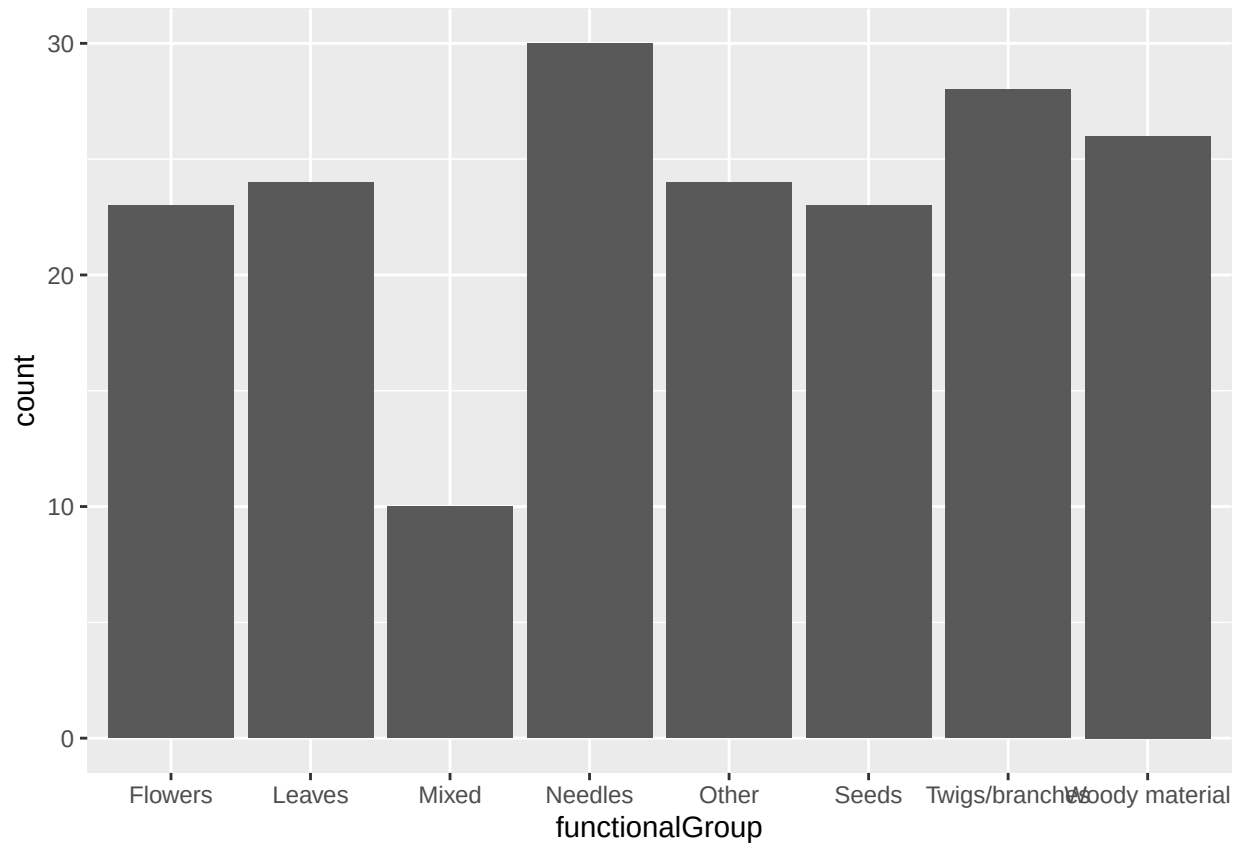
```
#listing plotIDs with summary function to compare
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14      8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: The `unique` function only provides a list of the plotIDs as factor levels, while the `summary` function provides all the PlotIDs along with the number of datapoints within each ID.

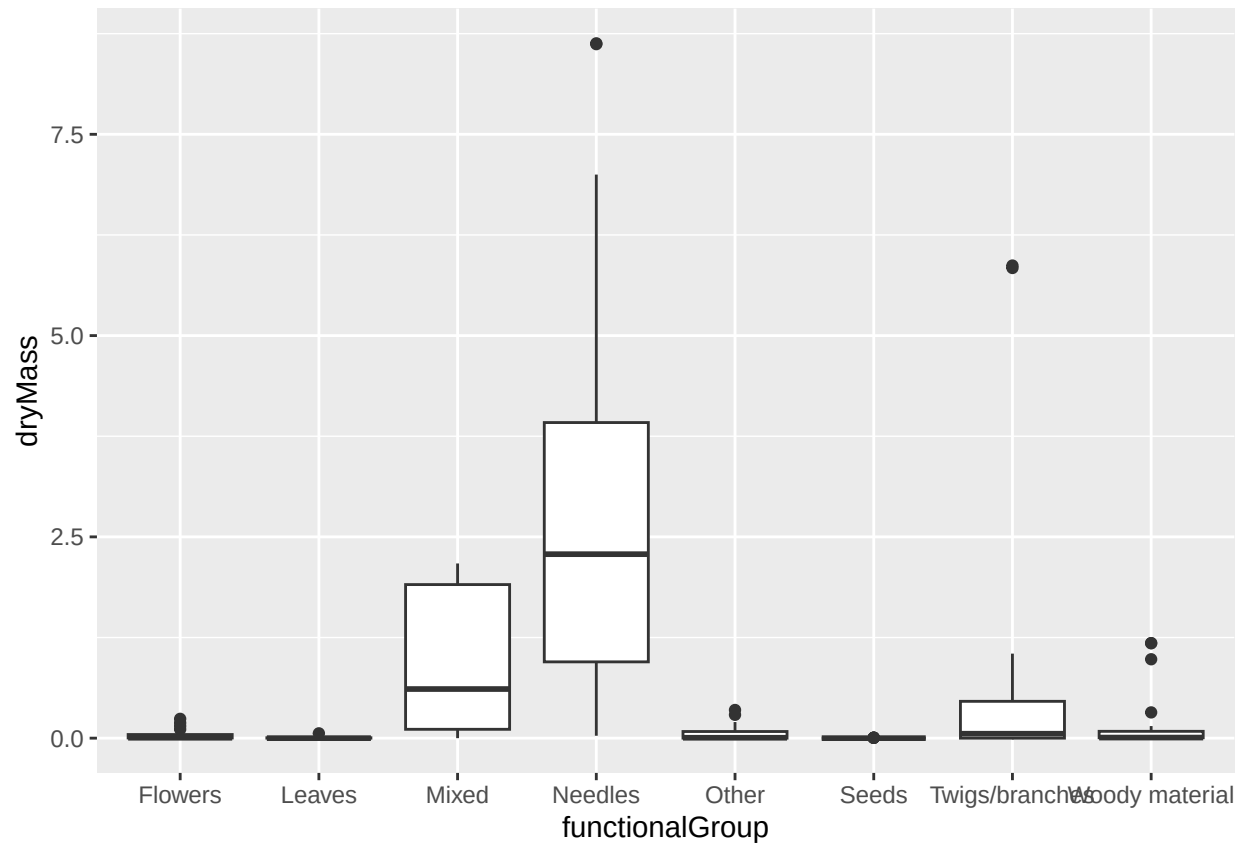
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
#bar graph of functional group counts
ggplot(Litter, aes(functionalGroup)) + geom_bar()
```

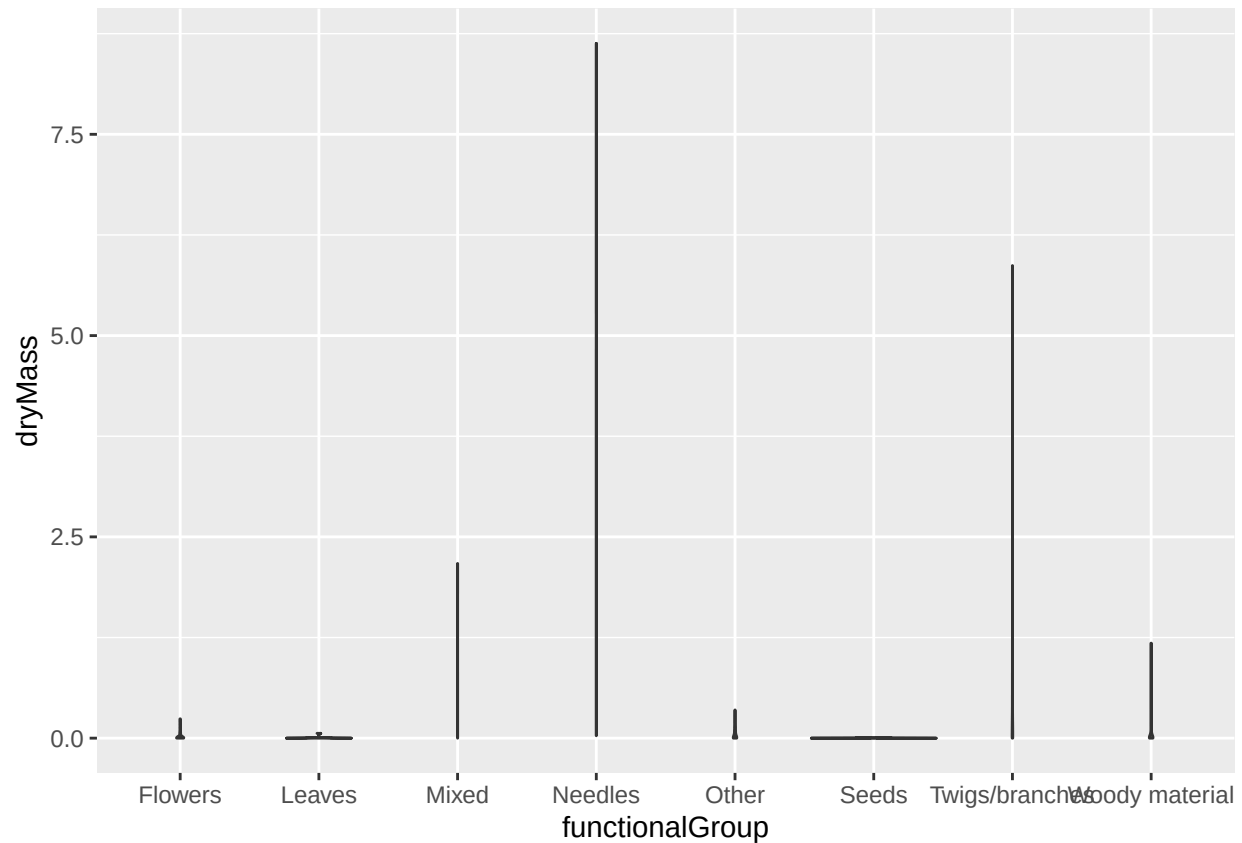


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
#boxplot  
ggplot(Litter, aes(functionalGroup, dryMass)) + geom_boxplot()
```



```
#violin plot  
ggplot(Litter, aes(functionalGroup, dryMass)) + geom_violin()
```



#observing the amount of points in each functional group
`summary(Litter$functionalGroup)`

```
##      Flowers      Leaves      Mixed      Needles      Other
##         23         24         10         30         24
## Seeds Twigs/branches Woody material
##         23         28         26
```

Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The different functional groups all have similar amounts of datapoints, meaning the density aspect of the violin plot is not legible for this dataset, and instead you only see vertical lines with little information.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles and twigs/branches. Mixed litters also have high biomasses.